

Keywords: reinforcement learning • zero RL • emergent reasoning • trillion-parameter scaling

Ring-Zero

Ant Group Scales Zero RL to One Trillion Parameters, Observing Five Emergent Reasoning Behaviors and 84.2% on AIME 2026 Without Human-Annotated Data

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Zero reinforcement learning — RL with verifiable rewards and no human-annotated data — has demonstrated striking results at the 70B scale, but whether its training dynamics and emergent capabilities extend to far larger models remained unknown. Ant Group’s InclusionAI now closes that gap: Ring-Zero applies zero RL to Ring-2.5-1T, a one-trillion-parameter hybrid linear-attention model, and documents not only successful scaling but five qualitatively new behaviors that appear only at this scale.

AT A GLANCE

Problem: Naive scaling of zero RL collapses beyond ~70B parameters due to readability degradation, token redundancy, and distribution drift.

Why it matters: Trillion-parameter reasoning models are now practical; understanding and controlling zero RL at that scale is necessary for continued progress.

Method: Clipped importance sampling, training-inference ratio correction, and mixed-precision control applied to Ring-2.5-1T (1T parameters, 1M context window).

Result: 84.2% on AIME 2026 with first-stage training only; five new emergent behaviors including self-verification and context anxiety.

Evidence: Evaluated on AIME 2024/2025/2026, HMMT, and IMOAnswerBench; seven-benchmark average 74.8%.

The Big Picture

Zero RL uses only the binary signal of whether the model’s answer is correct to teach chain-of-thought reasoning — no annotated thought chains, no human preference data, no teacher demonstrations. DeepSeek-R1-Zero showed that this surprisingly minimal training signal is sufficient to produce sophisticated mathematical reasoning at the 70B scale, but every subsequent study has operated at or below that size.

The trillion-parameter regime matters because it is where qualitatively new capabilities have historically emerged: GPT-4-class reasoning, multimodal integration, and in-context meta-learning all appeared near or above the 1T parameter threshold. Ring-Zero asks whether the *training dynamics* of zero RL also exhibit threshold behavior at this scale — and finds that they do.

Why Existing Approaches Fall Short

Three failure modes obstruct naive trillion-parameter zero RL. First, **poor readability**: at large scale, unconstrained policy gradient updates cause the model to generate verbose, repetitive internal monologues that consume tokens without reasoning. Second, **token redundancy**: the model learns to pad reasoning traces with low-effort repetition, inflating apparent solution length while contributing no additional correctness. Third, **distribution drift**: at 1T parameters, rollout distributions shift rapidly relative to the behavior policy used for importance-weighted gradient estimation, destabilizing training after a few hundred steps.

Existing stabilization techniques developed for 70B models (gradient clipping, KL penalties) are insufficient at 1T because the optimizer’s effective learning rate — after importance-weighted corrections — can grow dramatically when policy entropy is high in a trillion-parameter space.

Core Mechanism

Ring-Zero addresses the three failure modes with three targeted interventions. **Clipped importance sampling** restricts the off-policy correction ratio to $[1 - \epsilon, 1 + \epsilon]$ per token, preventing any single high-variance rollout from dominating the gradient update. **Training-inference ratio correction** compensates for the mismatch between the truncated context windows used during RL rollouts and the full context available at inference. **Mixed-precision control** uses BF16 for activations and FP32 for gradient accumulation, with selective FP8 for embedding layers to keep the 1T model within GPU memory constraints.

Technical Formulation

The training-inference ratio correction scales the effective reward by the ratio of inference to training context lengths:

$$\hat{r}_t = r_t \cdot \left(\frac{C_{\text{inf}}}{C_{\text{train}}} \right)^\gamma$$

where C_{inf} is the inference context length, C_{train} is the rollout context length, and γ is a hyperparameter tuned per model size (set to 0.4 for Ring-2.5-1T).

Clipped importance sampling modifies the standard im-

portance weight:

$$\tilde{\rho}_t = \min\left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 + \epsilon\right)$$

without a lower clip (since the lower clip would suppress on-policy updates when the policy has improved).

Learning Procedure

Training reliably unfolds in two sequential phases identified across multiple random seeds:

1. **Discovery phase** (first $\approx 30\%$ of steps): The model actively expands reasoning boundaries, activating dormant pathway combinations that were not used in pre-training. Output diversity is high; format is unstable.
2. **Sharpening phase** (remaining $\approx 70\%$ of steps): The model refines its policy within the stylistic and procedural boundaries established in discovery. Output quality and format consistency improve monotonically.

The phase transition is sharp and reproducible: gradient entropy drops by approximately 40% between steps 800 and 1200 across all runs, marking the boundary between discovery and sharpening.

Five Emergent Behaviors

Five behaviors appear at trillion-parameter scale that are absent at 70B, identified through systematic activation probing and output pattern analysis:

1. **Self-verification:** The model spontaneously checks intermediate conclusions before committing to a final answer, inserting verification sub-problems without being prompted to.
2. **Context anxiety:** The model models itself as a resource-constrained agent operating within a finite context window, making strategic decisions about what to include based on this self-model.
3. **Anthropomorphism:** The model adopts a persistent first-person problem-solving persona and maintains it across long reasoning chains.
4. **Structured formatting:** Headers, numbered steps, and boxed answers emerge without any formatting reward.
5. **Parallel reasoning:** The model explores multiple solution strategies in parallel within a single forward pass, then selects among them.

Experimental Findings

Benchmark	Ring-2.5-1T-Zero	Prior SOTA
AIME 2026	84.2%	78.5%
AIME 2025	81.7%	79.3%
AIME 2024	83.4%	82.1%
HMMT	78.3%	74.0%
IMOAnswerBench	61.2%	55.7%
7-Benchmark Avg.	74.8%	70.3%

Results are vendor-reported by InclusionAI; independent third-party evaluation was not yet available at press time.

Ablations and Interpretation

Without clipped importance sampling: Training diverges after approximately 500 steps at 1T scale; the effect is absent at 70B, confirming it is a trillion-parameter-specific failure mode.

Without training-inference ratio correction: AIME 2026 accuracy drops from 84.2% to 79.1%, a 5.1 percentage point loss attributable to underestimating long-context performance during training.

Context anxiety phase transition: The behavior is absent below $\sim 800\text{B}$ parameters and strengthens above 1T, suggesting a phase transition rather than a smooth scaling curve.

Self-verification contribution: Suppressing self-verification tokens at inference reduces AIME 2026 accuracy by approximately 2–3 percentage points, confirming its functional role in correctness.

Reference

Xinyu Tang et al. **Ring-Zero: Scaling Zero RL to a Trillion Parameters for Emergent Reasoning.** arXiv:2607.12395, July 2026. <https://arxiv.org/abs/2607.12395>

Keywords: unified vision • RGB encoding • dense perception • zero-shot transfer

RINO

A Multi-Institution Team Proposes Treating All Vision Tasks as RGB-to-RGB Image Editing, Enabling One Backbone to Handle Depth, Segmentation, and Normals Without Fine-Tuning

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The dominant paradigm in computer vision maintains separate models for separate tasks: one model for depth, another for segmentation, a third for surface normals. Even modern unified models typically require task-specific heads or adapters and must be fine-tuned per benchmark. RINO (RGB In, RGB Out) proposes a different architecture: encode every visual output as an RGB image and reduce every vision task to an RGB-to-RGB image editing problem, sharing all parameters across all tasks without task-specific components.

AT A GLANCE

Problem: Dense visual perception requires separate models per task; existing unified approaches still use task-specific heads or require benchmark fine-tuning.

Why it matters: True generalist vision requires a single set of parameters that transfers across tasks through a unified interface, as text does for NLP.

Method: All visual modalities (depth, masks, normals, keypoints) encoded as RGB images; tasks become text-conditioned RGB-to-RGB editing problems.

Result: Competitive zero-shot performance on four dense perception benchmarks using a single generic image editing backbone.

Evidence: NYU-Depth-v2, ADE20K segmentation, ScanNet normals, H3WB keypoints; zero-shot, no benchmark fine-tuning.

The Big Picture

Language models achieve generality through a single universal representation: text. Any NLP task — translation, summarization, QA, classification — can be expressed as a text-in, text-out problem, allowing one set of parameters to transfer across all of them. RINO asks whether pixels can serve the same function for vision.

The central observation is that all dense vision outputs are already spatial arrays over the same 2D grid as the input image. Depth is a scalar per pixel, surface normals are a 3-vector per pixel, segmentation masks are a class label per pixel, and keypoint heatmaps are a confidence per pixel. All of these can be rendered as an RGB image and decoded back without loss, if the encoding is chosen carefully.

Why Existing Approaches Fall Short

Task-specific models are powerful but expensive: each benchmark requires a separate model, separate fine-tuning run, and separate inference process. Multi-task models (e.g., UniPerceiver, InternImage) unify multiple tasks in one architecture but still require task-specific output heads that must be trained on benchmark data. Generalist models like Painter and PromptDiffusion reduce fine-tuning requirements but retain task-specific structured output formats.

The key gap is that no prior work treats the task output itself as RGB data fed into a backbone that was not designed for the task at all. RINO demonstrates that a backbone trained entirely on natural image editing — with no exposure to depth maps, segmentation masks, or surface normals — can perform these tasks zero-shot if the output is presented to it as an RGB image to edit.

Core Mechanism

Each visual modality is assigned a lossless or near-lossless RGB encoding:

- Depth maps:** Depth values d are normalized to $[0, 255]$ and stored in the red channel. The green channel encodes epistemic uncertainty; blue is zero.
- Segmentation masks:** Each semantic class receives a fixed, perceptually distinct RGB color. Instance masks use hue-shifted variants of the class color, enabling visual discrimination of instances.
- Surface normals:** The XYZ components of the unit normal vector are linearly mapped to $[0, 255]$ triplets: $\text{RGB} = \lfloor 127.5(\mathbf{n} + \mathbf{1}) \rfloor$.
- Keypoint heatmaps:** Per-joint Gaussian confidence heatmaps are stacked into the RGB channels (up to three joints per RGB image, with multiple images for larger joint sets).

A text instruction specifies the desired output type. The editing backbone processes the input image and instruction, producing an RGB output that is decoded into the task-specific representation.

Technical Formulation

For a task T with input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ and ground truth output $\mathbf{y}_T$, RINO defines an encoding function $\phi_T : \mathbf{y}_T \rightarrow \mathbb{R}^{H \times W \times 3}$ and decoding function ϕ_T^{-1} . The backbone f_θ is trained to minimize:

$$\mathcal{L} = \mathbb{E}_{(x, y_T, T)} \|f_\theta(\mathbf{x}, \tau_T) - \phi_T(\mathbf{y}_T)\|_2^2$$

where τ_T is the text instruction for task T . Crucially, f_θ is the pretrained image editing backbone — its weights are frozen for zero-shot evaluation.

Experimental Findings

Task	Metric	RINO	Fine-Tuned
Depth (NYU-v2)	AbsRel↓	0.137	0.103
Seg. (ADE20K)	mIoU↑	38.7	47.2
Normals (ScanNet)	Mean Ang.↓	11.3°	9.1°
Keypoints (H3WB)	PCK↑	61.4	68.9

RINO’s zero-shot numbers are within 15–25% (relative) of dedicated fine-tuned models on each task, using a single set of backbone weights that was never trained on any of these datasets.

Ablations and Interpretation

Task-specific heads: Adding a lightweight task-specific head (while keeping the backbone frozen) closes approximately half the gap to fully fine-tuned models on depth and segmentation. This confirms that the backbone captures the necessary spatial representations; the remaining gap is in task-specific prediction.

RGB encoding quality: Using a uniform grayscale depth encoding (instead of the normalized red-channel scheme) degrades depth AbsRel by 0.021 absolute, confirming that encoding choices matter substantially.

Instruction specificity: Vague instructions (“transform this image”) yield near-random outputs; specific instructions (“generate a surface normal map with outward normals colored red-green-blue”) approach zero-shot quality, suggesting the backbone uses the instruction as a strong conditioning signal.

Joint training benefit: When the backbone is fine-tuned jointly across depth + normals + segmentation (vs. separately), each task improves by 1–2%, suggesting mutual regularization from shared RGB representations.

Reference

Timing Yang et al. **Let RGB Be the Language of Vision**. arXiv:2607.12450, July 2026. <https://arxiv.org/abs/2607.12450>

Keywords: hallucination • parametric knowledge • evidence grounding • pretraining

Knowledgeless Language Models

Anonymizing Named Entities at Pretraining Time Suppresses Parametric Factual Recall While Improving Contextual QA, Fact Verification, and Hallucination Detection

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Language models encode factual knowledge in their parameters during pretraining, and this parametric knowledge is a double-edged sword. It enables impressive zero-shot question answering but also produces confident hallucinations when the model’s memorized facts are outdated, incomplete, or contradicted by the provided context. KLLMs (Knowledgeless Language Models) address this at the source: instead of correcting parametric recall through fine-tuning or prompting, they prevent it from forming in the first place by anonymizing named entities before pretraining begins.

AT A GLANCE

Problem: LLMs memorize entity-linked facts during pretraining and hallucinate them confidently when context contradicts or omits them.

Why it matters: Hallucination and parametric override are root causes of LLM unreliability in knowledge-intensive tasks; existing mitigations act at fine-tuning or inference time.

Method: Named entities in the pretraining corpus are replaced with consistent-within-document, random-across-documents placeholder tokens, blocking entity-linked factual supervision.

Result: +12.2% relative on open-book NaturalQuestions; +8.5% on FEVER; +9.3% on HaluEval; near-zero perplexity cost.

Evidence: 125M, 350M, and 1.3B model scales; five benchmarks spanning QA, fact verification, and hallucination detection.

The Big Picture

When a language model is asked “Who directed Oppenheimer?” without any context, it answers from its parameters. When it is asked the same question with a context passage, it should answer from that passage — but it often

does not, especially when the parametric answer conflicts with the contextual one.

This *parametric override* problem is distinct from the out-of-distribution problem: it occurs even when the model has seen the correct answer in context. The issue is that parametric knowledge has been learned so strongly during pretraining that it dominates contextual signals in downstream deployment.

KLLMs test a simple hypothesis: if named entities are removed from pretraining data (replaced by anonymized placeholders), the model cannot form entity-specific parametric associations in the first place, making it structurally dependent on context for entity-linked facts.

Why Existing Approaches Fall Short

Retrieval-augmented generation (RAG) mitigates hallucination at inference time but does not change the model’s internal representation or its tendency to override retrieved context with parametric beliefs. Fine-tuning with contrastive or knowledge-editing objectives can suppress specific facts but does not address the pretraining-level memorization mechanism.

Prompting strategies (“base your answer only on the passage”) improve context use in controlled settings but fail in adversarial cases where the model’s high-confidence parametric recall outweighs the instruction. KLLMs target the fundamental source: the pretraining signal that creates entity-linked parametric knowledge.

Core Mechanism

A named entity recognition (NER) model processes each document and identifies spans corresponding to persons, organizations, locations, dates, and named products. Each unique entity within a document is replaced with a type-annotated placeholder: [PERSON_42], [ORG_7], etc.

Critically, the mapping from entities to placeholders resets between documents. The same person gets placeholder [PERSON_42] in one document and [PERSON_19] in another. This prevents the model from learning cross-document entity associations — the mechanism by which parametric factual knowledge accumulates.

Within a document, coreference is preserved: if [PERSON_42] is the subject of the first sentence, it remains [PERSON_42] throughout, so the model still learns intra-document discourse structure. Linguistic syntax, semantic roles, and factual relations within each document are fully intact; only entity identity across documents is suppressed.

Technical Formulation

Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be the pretraining corpus. Let $E(d_i) = \{e_1, \dots, e_k\}$ be the set of unique entities in document d_i . The anonymized document is:

$$\tilde{d}_i = \sigma_i(d_i)$$

where σ_i is a randomly sampled bijection from entities in d_i to type-specific placeholder tokens, drawn fresh for each document. The KLLM is trained with the standard autoregressive cross-entropy loss on $\{\tilde{d}_i\}$; no special objective is needed.

The cross-document anonymization satisfies: for any two documents $d_i \neq d_j$ and any entity e , with probability 1 the placeholder for e in $\tilde{d}_i$ differs from its placeholder in $\tilde{d}_j$.

Experimental Findings

Benchmark	KLLM-1.3B	Baseline-1.3B
NQ open-book EM $\uparrow$	43.1%	38.4%
NQ closed-book EM $\uparrow$	6.2%	18.7%
FEVER accuracy $\uparrow$	83.7%	79.2%
HaluEval $\uparrow$	72.4%	65.1%
TruthfulQA $\uparrow$	48.3%	41.7%
Perplexity $\downarrow$	14.3	14.1

The closed-book NQ drop (18.7% $\rightarrow$ 6.2%) confirms that entity anonymization successfully suppresses parametric factual recall. The open-book NQ gain (+12.2% relative) confirms that the model compensates by better utilizing context. The perplexity difference of 0.2 points (1.4% relative) confirms that linguistic competence is essentially unaffected.

Ablations and Interpretation

Partial anonymization: Anonymizing only person entities (not organizations or locations) yields approximately half the HaluEval improvement, suggesting that all entity types contribute to the parametric recall problem.

Fixed vs. random cross-document mapping: Using a fixed entity-to-placeholder mapping across all documents partially re-introduces factual associations (the model can form cross-document entity statistics by token co-occurrence). The random-per-document approach is essential for full suppression.

Downstream fine-tuning: KLLMs fine-tuned on Wikipedia-based QA recover most of the closed-book performance while retaining the open-book advantage, suggesting that the intervention targets pretraining memorization specifically rather than the model’s capacity to learn facts at all.

Scale: The open-book improvement grows from +6.1% at 125M to +12.2% at 1.3B, suggesting that larger KLLMs

are better at leveraging context when parametric recall is suppressed.

Reference

Roi Cohen et al. **Knowledgeless Language Models: Suppressing Parametric Recall for Evidence-Grounded Language Modeling**. arXiv:2607.12831, July 2026. <https://arxiv.org/abs/2607.12831>

Keywords: credit assignment • long-horizon agents • reinforcement learning • tool use

TRACE

Microsoft Research Derives Dense Turn-Level Rewards for Long-Horizon Agents from Temporal-Difference Changes in Frozen Reference Model Log-Probabilities

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Multi-turn tool-use agents have emerged as a dominant paradigm for complex task completion: the agent writes and executes code, calls APIs, browses the web, edits files, and iterates over dozens to hundreds of steps before producing a final answer. Outcome-only reinforcement learning provides a single binary signal at the end of this trajectory. TRACE (Turn-level Reward Assignment via Credit Estimation) replaces this single sparse signal with dense, turn-by-turn rewards derived from how each tool call changes the model’s probability of reaching the correct answer — without any human annotation of intermediate steps.

AT A GLANCE

Problem: Outcome rewards become sparse, high-variance, and misleading for agents operating over 20–100 tool-call trajectories.

Why it matters: Long-horizon agents are increasingly deployed for software engineering, scientific computing, and complex information retrieval; outcome-only RL trains them poorly.

Method: State values estimated as log-ratio of gold-answer log-probabilities (current vs. frozen reference); turn-level rewards are TD differences between consecutive state values.

Result: ToolBench task completion: 41.3% → 58.7%; WebArena: 19.4% → 28.1%; long-horizon code repair: 31.6% → 47.2%.

Evidence: ToolBench, WebArena, and a 20-step code repair benchmark; no intermediate human annotations used.

The Big Picture

Reinforcement learning for single-turn tasks is straightforward: an answer is either right or wrong, and the reward is binary. For multi-turn agents, the binary outcome arrives only at the end of a trajectory that may span dozens of

tool calls and intermediate reasoning steps. Any single turn — a web search, a code edit, a database query — affects the probability of the final correct answer, but outcome-only RL provides no per-turn signal about which turns were helpful.

The credit assignment problem is further complicated by the misleading nature of failed trajectories: a sequence of mostly correct intermediate steps followed by a single error at the last step receives the same negative reward as a trajectory that was wrong from the start, even though the former’s intermediate actions should be reinforced.

Why Existing Approaches Fall Short

Monte Carlo rollout-based credit assigns turn-level rewards by averaging outcomes over multiple rollouts starting from each intermediate state. This is statistically sound but computationally prohibitive for long trajectories: estimating a reliable MC signal for each of 50 turns in a trajectory requires $50 \times K$ rollouts per training example.

Process reward models (PRMs) learn to score intermediate steps, but require human annotation of intermediate step quality, which is expensive and does not scale to the diverse tool-use environments where agents are deployed.

Hindsight relabeling and advantage-based credit methods improve over uniform assignment but remain high-variance for very long trajectories because the value function must extrapolate many steps into the future.

Core Mechanism

TRACE introduces a principled value function for multi-turn trajectories that requires no additional training: the log-ratio of the gold-answer log-probability under the current policy versus a frozen reference model.

At each state s_t (the trajectory prefix through turn t), the state value is:

$$V(s_t) = \log \frac{p_\theta(y^* | s_t)}{p_{\text{ref}}(y^* | s_t)}$$

where y^* is the known gold answer. This quantity has a natural interpretation: it measures how much more likely the current policy is (relative to the frozen reference) to produce the correct answer given the information available at turn t .

The turn-level reward is the Temporal-Difference change:

$$r_t = V(s_t) - V(s_{t-1})$$

Technical Formulation

A trajectory of T turns decomposes into state transitions: $(s_0, a_1, s_1, a_2, \dots, a_T, s_T)$, where each action a_t is a tool call and s_t is the trajectory prefix after the tool result.

The policy π_θ maps from s_{t-1} to a_t ; the tool environment is deterministic.

The TRACE reward satisfies a telescoping property:

$$\sum_{t=1}^T r_t = V(s_T) - V(s_0)$$

where $V(s_T)$ approximates the outcome reward and $V(s_0)$ is a constant baseline. Thus TRACE distributes the outcome signal across turns in a way that is globally consistent: turns that increase V receive positive rewards, turns that decrease it receive negative rewards, and the sum equals the net outcome advantage.

Log-probabilities $p_\theta(y^* \mid s_t)$ are computed by teacher-forcing the gold answer given the trajectory prefix, a standard operation for autoregressive models that requires one forward pass per turn.

Experimental Findings

Benchmark	Outcome RL	TRACE RL
ToolBench	41.3%	58.7%
WebArena	19.4%	28.1%
Code Repair (>20 steps)	31.6%	47.2%

Credit quality measured against human annotations (Pearson r): TRACE achieves $r = 0.61$ vs. $r = 0.22$ for MC rollout-based credit and $r = 0.09$ for uniform assignment.

Ablations and Interpretation

Uniform credit (assigning $V(s_T)/T$ to each turn): Performance degrades by 8–12% across benchmarks, confirming that the temporal structure of the TD signal is informative — not just the magnitude of the outcome.

Reference model choice: Using a weaker (smaller) reference model degrades credit quality, suggesting the value of a calibrated log-probability baseline. Using the same model checkpoint before RL (as TRACE prescribes) consistently outperforms alternatives.

Trajectory length: The TRACE advantage grows with trajectory length. For tasks with fewer than 5 tool calls, the improvement over outcome-only RL is <2 percentage points. For tasks with 20+ tool calls, the gap is 15+ percentage points, consistent with the hypothesis that credit assignment difficulty scales with trajectory length.

Negative turn identification: TRACE correctly assigns negative rewards to turns that move the agent away from the goal — detectable as a decrease in state value — enabling the policy to specifically learn to avoid those actions.

Reference

Leitian Tao et al. **TRACE: Turn-level Reward Assignment via Credit Estimation for Long-Horizon Agents**. arXiv:2607.13988, July 2026. <https://arxiv.org/abs/2607.13988>

Keywords: diffusion language models • policy gradient • masked diffusion • RLHF

Mask-Aware Policy Gradients

UT Austin Corrects a Fundamental Bias in RL Training for Masked Diffusion Language Models by Properly Accounting for Both Token and Masking Decisions

By Haran Raaresh, Kulin Shah, Adam Klivans, Philipp Krähenbühl (University of Texas at Austin)

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Masked Diffusion Language Models (MDLMs) generate text by iteratively unmasking tokens over multiple steps. Applying reinforcement learning to MDLMs — to improve performance on math reasoning, code generation, or instruction following — requires a valid policy gradient. But all existing RL approaches for MDLMs compute a policy gradient that ignores one of the model’s two fundamental decisions: not just which tokens to predict, but which positions to remask at each step. TRACE-MDLM corrects this omission with a two-stage action MDP formulation, raising GSM8K accuracy from 79.3% to 87.1% and MBPP from 41.2% to 53.4%.

AT A GLANCE

Problem: RL for MDLMs ignores the masking schedule decision in the policy gradient, producing a biased update that misattributes credit to token choices.

Why it matters: MDLMs are an increasingly competitive alternative to autoregressive LMs; reliable RL is necessary for closing the remaining gap to AR models on reasoning tasks.

Method: Two-stage action MDP: the policy gradient decomposes into a token term and a masking term; a lightweight MLP head is trained to parameterize the masking policy.

Result: 87.1% on GSM8K (+7.8 pp over token-only baseline); 53.4% on MBPP (+12.2 pp); 40% lower gradient variance.

Evidence: Evaluated on GSM8K and MBPP; ablations on 3B-scale MDLM; accepted at COLM 2026.

The Big Picture

Masked Diffusion Language Models are a promising alternative to autoregressive (AR) LMs: they are intrinsically parallelizable across token positions, support flexible bidirectional context, and can be sampled at adjustable quality-speed tradeoffs by varying the number of denoising steps. MDLMs have closed much of the gap to AR

models on language modeling benchmarks, but lag behind on structured reasoning tasks like mathematics and code.

Reinforcement learning from verifiable rewards — the technique that drove major gains for AR reasoning models via GRPO, PPO, and similar methods — is the natural next step for MDLMs. The obstacle is that naive application of AR policy gradient methods to MDLMs produces a biased estimator, because it ignores a decision that is structurally part of the MDLM generation process: the masking schedule.

Why Existing Approaches Fall Short

Standard RL for AR models computes the log-probability of each generated token and uses these log-probabilities directly in the policy gradient. For MDLMs, the analogous computation scores each predicted token at each unmasked position — ignoring that the model also decided which positions to keep unmasked (and which to remask) at each step.

The masking schedule is not a fixed algorithmic artifact: it is a learned policy decision. A given sequence of T denoising steps involves T masking decisions, each dependent on the model’s output and the task. Ignoring these decisions is equivalent to computing the policy gradient for an AR model while ignoring every other token in the generation — the gradient is biased and the resulting training signal miscredits token choices for outcomes that were actually driven by masking decisions.

Prior approaches including SPG and GRPO-for-MDLMs apply variance reduction techniques to the token gradient without addressing this structural bias.

Core Mechanism

The paper formalizes MDLM generation as a two-stage MDP per denoising step. At step t , the model observes a partially masked sequence $\mathbf{x}_t$ and takes two sequential actions:

1. **Token action** a_t^{tok} : For each masked position $i \in \mathcal{M}_t$, predict the unmasked token x_i from the MDLM network. This recovers a fully unmasked intermediate sequence $\hat{\mathbf{x}}_t$.
2. **Masking action** a_t^{mask} : Select a subset of positions in $\hat{\mathbf{x}}_t$ to remask, producing $\mathbf{x}_{t+1}$ for the next step. This is parameterized by a lightweight MLP over the MDLM hidden states.

The full policy gradient decomposes as:

$$\nabla_{\theta} J = \mathbb{E}_{\pi} \left[\hat{A}_t (\nabla_{\theta} \log \pi_{\theta}(a_t^{\text{tok}} \mid \mathbf{x}_t) + \nabla_{\theta} \log \pi_{\theta}(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t)) \right]$$

where $\hat{A}_t$ is the advantage estimate at step t .

Technical Formulation

The masking policy $\pi_\theta(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t)$ is a factored Bernoulli over positions:

$$\pi_\theta(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t) = \prod_{i=1}^L \text{Bernoulli}(m_i; \sigma(\mathbf{W}\mathbf{h}_i + b))$$

where $\mathbf{h}_i$ is the MDLM hidden state at position i after the token action, $\mathbf{W} \in \mathbb{R}^{1 \times d}$ and $b \in \mathbb{R}$ are the MLP parameters, and $m_i \in \{0, 1\}$ is the masking decision. The MLP is trained jointly with the denoising network.

The advantage $\hat{A}_t$ is estimated using Generalized Advantage Estimation (GAE) with $\lambda = 0.95$ and a value function head trained on the MDLM encoder representations.

Experimental Findings

Method	GSM8K	MBPP
MDLM-3B (no RL)	52.1%	28.7%
Token-only REINFORCE	79.3%	41.2%
SPG	81.4%	44.7%
Mask-Aware PG (ours)	87.1%	53.4%

The masking term alone contributes approximately 3–4 percentage points; jointly optimizing both token and masking gradients provides the full 7–12 percentage point improvement over token-only methods.

Gradient norm variance is 40% lower with Mask-Aware PG, explaining the smoother training curves and reduced need for gradient clipping.

Ablations and Interpretation

Token gradient only: Removing the masking term reproduces the SPG/REINFORCE baseline, confirming the masking term as the source of improvement.

Masking term only: Training with only the masking gradient (no token gradient) fails to converge, confirming both terms are jointly necessary.

Masking schedule analysis: Models trained with Mask-Aware PG learn to unmask tokens in a semantically coherent order: in math problems, nouns and verbs are unmasked first, modifiers last. This learned curriculum differs qualitatively from the uniform-random schedule used at initialization.

Scaling: The advantage grows with problem difficulty. For easy single-step GSM8K problems the gap is <3 percentage points; for multi-step problems requiring 4+ intermediate computations the gap is 8+ percentage points, consistent with the masking schedule being most important when sequential reasoning structure matters.

Reference

Haran Raajesh, Kulin Shah, Adam Klivans, Philipp Krähenbühl. **Mask-Aware Policy Gradients for Diffusion Language Models.** COLM 2026 / arXiv:2607.15200, July 2026. <https://arxiv.org/abs/2607.15200>